“Stop Trying to Make Fetch Happen”: Towards a Modality-Aware Understanding of Unnecessary Resources on the Web

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IMC 2026

Compared to ~1.5MB ten years ago

~3MB

76 requests

median webpage, HTTP Archive, 2026

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Much of it unneeded for automated agents, archivers, crawlers

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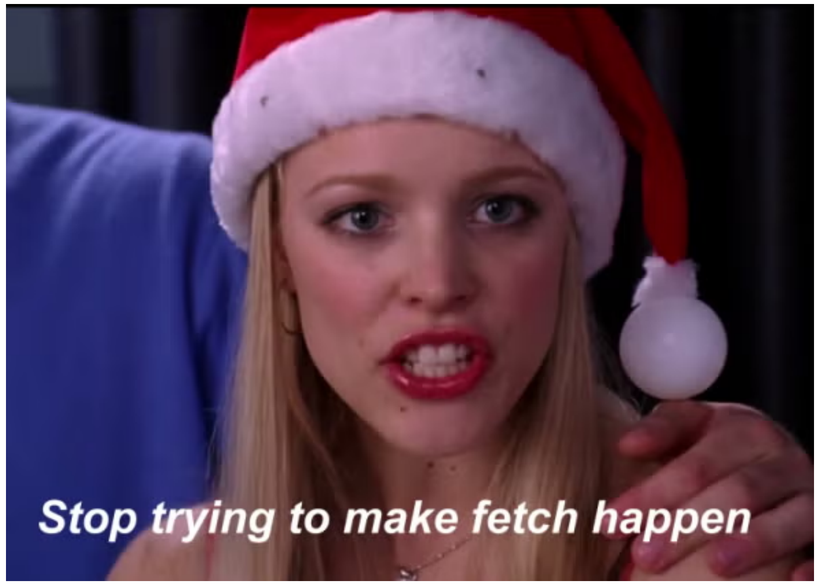
Yet they all fetch the same bundle of requests

Why does overfetching matter?

- Concern over the impact of the web on the environment (Sustainability)
- Cost of web access to increasingly complex websites (Affordability)

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Prior work

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- Unused JavaScript

Our work

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Necessity isn’t a property of the page; it depends on who’s asking

Our work

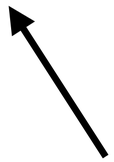
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- Depends on fidelity requirements

Our work

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- Can include other resource types
- Depends on fidelity(?) requirements



Similarity of the page to the original

Dimensions of Fidelity

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Textual

Dimensions of Fidelity

Textual

Structural

Dimensions of Fidelity

Textual

Structural

Visual

Dimensions of Fidelity

Textual

Structural

Visual

Interactive

Dimensions of Fidelity

Textual

Main text content extracted with Mozilla Readability and compared with Jaccard string similarity

Structural

Weighted Jaccard similarity between extracted DOM nodes

Visual

SSIM on extracted components of the webpage (QSS)

Interactive

Weighted success rate of task attempts by an AI browser agent (BrowserUse)

Modalities

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Strict Fidelity

Textual

Structural

Visual

Interactive

Modalities

Strict Fidelity	Textual	Structural	Visual	Interactive
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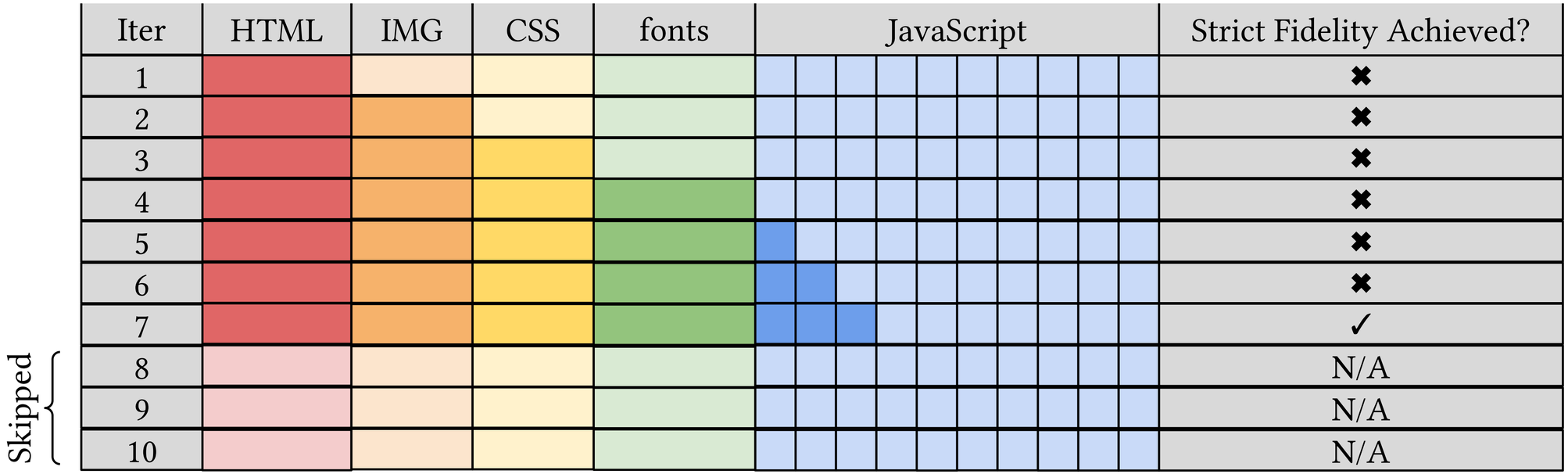
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Strict Fidelity	Textual	Structural	Visual	Interactive
Text Extraction	Textual	Structural	Visual	Interactive
Content Archival	Textual	Structural	Visual	Interactive
Programmatic Automation	Textual	Structural	Visual	Interactive
Human Browsing	Textual	Structural	Visual	Interactive

Does accounting for modality change how we perceive unnecessary resources on the web?

Measuring Resource Necessity

- Sampled buckets of the CrUX global top sites
  - Total 4000 domains
- Loaded each webpage twice to get a baseline score for perfect fidelity
- Reloaded each webpage while incrementally re-enabling resource categories
  - Re-enabled external JS one script at a time


Iter
HTML
IMG
CSS
fonts
JavaScript
Strict Fidelity Achieved?
1
2
3
4
5
6
7
8
9
10
✖
✖
✖
✖
✖
✖
✓
N/A
N/A
N/A
Skipped

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Not the optimal solution, one reasonable loading strategy

Findings

1. Modalities reveal additional bloat
2. Resources needed to preserve fidelity
3. Why reductions fail
4. Bloat concentration
5. Correlation between measures of fidelity

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- If a more restrictive modality is defined, this number jumps to 54–93%
  - With a median reduction of 17-98% of the page size
  - Content archival and text extraction on the opposite ends

Modalities reveal additional bloat contd.

Using the Sustainable Web Design Model (SWDM)

Modalities reveal additional bloat contd.

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If we expect strict fidelity:

86.5 million
metric tons of CO_2e per year

Context:
Driving a car for one mile in the US
results in 400g CO_2 emitted



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Modalities reveal additional bloat contd.

Using the Sustainable Web Design Model (SWDM)

Weighting by percentage of human and bot traffic and their associated unnecessary resources by modality:

159
8[illegible]5 million
metric tons of CO_2e per year

Context:
Driving a car for one mile in the US results in 400g CO_2 emitted

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Modalities reveal additional bloat contd.

Using the Sustainable Web Design Model (SWDM)

Carbon footprint of unnecessary resources is larger than we thought!

159 ~~8[illegible]5~~ million
metric tons of CO_2e per year

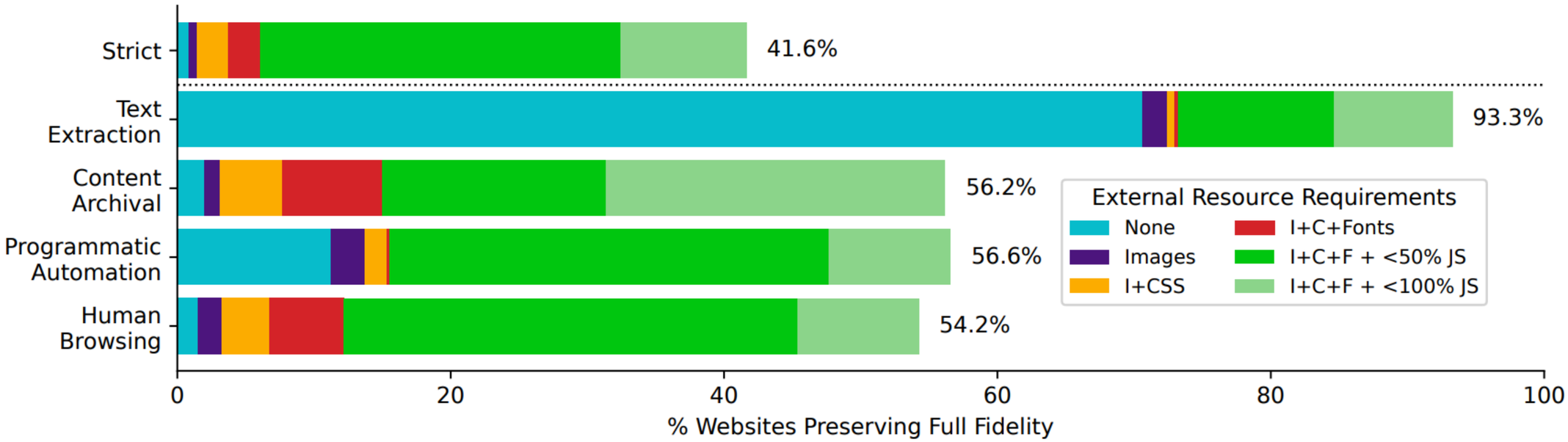
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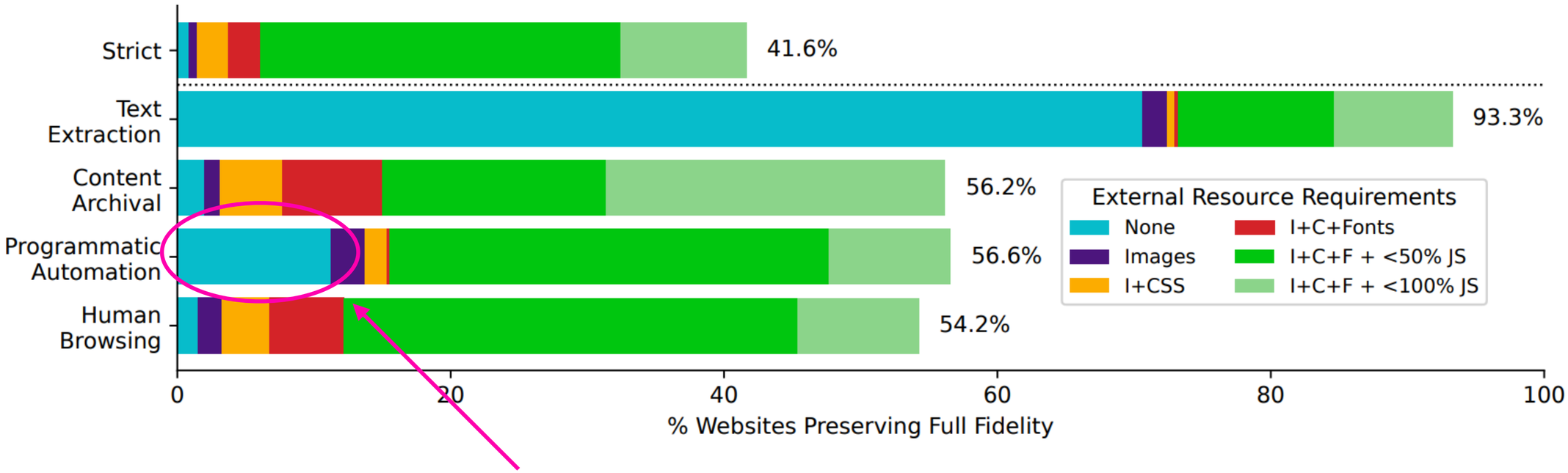
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Resources needed to preserve fidelity

- Many sites can be reduced a subset of their resources
- Some need no external resources at all
- Even when external JS is required, only a subset is sufficient


Strict
Text Extraction
Content Archival
Programmatic Automation
Human Browsing
41.6%
93.3%
56.2%
56.6%
54.2%
External Resource Requirements
None
Images
I+CSS
I+C+Fonts
I+C+F + <50% JS
I+C+F + <100% JS
0
20
40
60
80
100
% Websites Preserving Full Fidelity



Many webpages don't need external resources
if visual output is not necessary

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Fidelity is not a one dimensional concept!

Takeaways

Measure fidelity according to the relevant dimension as one doesn’t predict the others

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Optimize crawlers for required measures of fidelity

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Paper 📄

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